SentiFolio

Part B: Funds, Sentiment & App

LINK TO GITHUB: https://github.com/jia-xuan-z/z5555674_projectB

LINK TO STREAMLIT: https://z5555674projectb-eppv6ytyyx2jducyw9luh2.streamlit.app/

1. Funds and backtest design

1.1 Product architecture

SentiFolio offers fifteen investable funds, one for every combination of asset universe (equity, crypto, combined) and optimisation method (equal-weight, minimum-variance, maximum-Sharpe, risk parity, mean-CVaR). Equity funds contain 50 US large-cap stocks across ten sectors, crypto funds contain 10 cryptocurrencies, and combined funds place both groups in one opportunity set. Considering each (universe, method) pair as a separate fund allows investors to first choose the asset class, then the construction rule.

All fifteen funds are long-only and fully invested, so every dollar is allocated and no position is negative. This structure allows an investor to separate two decisions which are how much exposure to equities versus crypto, and how defensively or aggressively that exposure is built.

1.2 Optimisation methods

Each method turns the same two inputs, the estimated mean return vector μ and covariance matrix Σ over a trailing window, into a weight vector w. Equal weight ignores both inputs and is the naive 1/N benchmark. According to DeMiguel, Garlappi & Uppal (2009), no “clever” rule reliably beats it out of sample, which is exactly the comparison Section 2.1 reports.

$$w_i = 1/N.$$

Minimum-variance (Markowitz, 1952) sets the weights with the smallest portfolio variance:

$$\min_{w} \; w^T \Sigma w.$$

Maximum-Sharpe, or tangency (Sharpe, 1966), sets the weights with the best return per unit of risk, using the assumed risk-free rate r

$$\max_{w} \; \frac{w^T(\mu - r_f 1)}{\sqrt{w^T \Sigma w}}.$$

Risk parity (Maillard, Roncalli & Teiletche, 2010) gives every asset the same share of total portfolio risk, solving for w such that each asset's risk contribution RC

$$RC_i = \frac{w_i(\Sigma w)_i}{w^T \Sigma w} = 1/N \quad for\ every\ asset\ i.$$

Every method is solved subject to full investment and no short selling:

$$1^T w = 1, \ w \geq 0.$$

Mean-CVaR (Rockafellar & Uryasev, 2000) targets tail risk directly, minimising the average loss during the worst-performing (1−α) proportion of days in the trailing window while requiring the portfolio to meet a minimum expected-return target. Unlike Maximum Sharpe, which measures risk via overall return volatility, Mean-CVaR focuses specifically on severe downside outcomes. With daily returns R in the trailing window, the minimisation is exactly representable as the linear program below, using an auxiliary VaR estimate ζ and one shortfall variable per day, u:

$$\min_{w,\zeta,u} \ \zeta + \frac{1}{(1-\alpha)T} \sum_t u_t.$$

The program is subject to full investment and no short selling as above, a floor on expected return, and one pair of constraints per day t in the window:

$$w^T \mu \geq \bar{\mu}, \quad u_t \geq -R_t w - \zeta, \quad u_t \geq 0.$$

$\bar{\mu}$ is the average individual asset's own estimated mean return in the window. The rule is while minimising tail risk, do not give up more expected return than a typical single asset in the universe would offer. This is a principled floor, not an arbitrary tuning knob. The first attempt maximised (mean − CVaR ratio) directly with SLSQP, mirroring maximum-Sharpe's shape, but Section 1.4 explains why that failed and this linear programming formulation replaced it.

1.3 Walk-forward method and assumptions

All funds are backtested out of sample with a walk-forward design. At each monthly rebalancing date t, weights are estimated from the preceding 252 trading-day window, strictly excluding date t, so no weight is influenced by the return realised on the rebalancing date or by any future information. The estimated weights are then held fixed until the next rebalance.

The equity-only and combined funds begin their out-of-sample period on 4 January 2021. The crypto-only funds use the same 252-observation lookback, but because crypto trades every calendar day this span roughly 252 calendar days, so they begin on 1 October 2020. The risk-free rate is assumed zero. Equity and combined performance is annualised on 252 trading days, crypto on 365 to reflect continuous trading. Transaction costs, bid-ask spreads, taxes and management fees are assumed zero throughout, a simplification revisited in Section 6.2.

Realised performance is summarised with the annualised Sharpe ratio, using mean daily return R̄, its standard deviation σ, and the risk-free rate r

$$SR = \frac{R - r_f}{\sigma}.$$

and the maximum drawdown, the worst peak-to-trough fall in the growth of $1 series V over the whole out-of-sample period:

$$DD_t = \frac{V_t}{max_{s \le t} V_s} - 1.$$

1.4 Optimiser validation

An implementation issue surfaced during validation is although scipy's SLSQP solver returned success, the minimum-variance and risk-parity weights for several rebalances stayed exactly at 1/N. A single-rebalance test showed SLSQP stopping after one iteration because daily covariance objectives are tiny (about 1e-4) relative to the solver's default absolute tolerance, so the solver reported convergence without meaningfully searching.

Before correction, across the backtests of nine estimation methods, all 111 risk-parity rebalances and 31 of 36 equity minimum-variance rebalances were affected. Scaling the objective by a positive constant and tightening the solver tolerance does not change the mathematical optimum, but fixes the numerical search. After correction, only 3 of the 333 method estimated rebalances failed to converge, each falling back safely to equal weight for that rebalance. A successful solver status alone did not prove that meaningful optimisation had occurred.

Mean-CVaR exposed a second different convergence failure. An initial version maximised the mean-CVaR ratio with SLSQP and repeatedly failed to converge. 8 of 36 Equity, 2 of 39 Crypto, and 13 of 36 Combined rebalances.

The cause was different, so rescaling the objective does not apply here. CVaR is defined through an empirical quantile of the scenario returns, so the worst-tail scenario set is itself a function of w. Wherever a small change in w swaps one scenario into or out of that tail, the objective has a kink and is not differentiable. SLSQP assumes a smooth, differentiable objective, so the failure was structural, and the fix was to change the formulation instead of retuning the solver. Minimising CVaR subject to linear constraints is instead an exact linear program (Section 1.2), with no convergence risk of this kind. After reformulating, zero rebalances failed to converge across all three Mean-CVaR backtests.

2. Out-of-sample results and fund fact sheets

2.1 Performance across the fifteen base funds

Table 1 compares annualised return, volatility, Sharpe ratio, maximum drawdown, and the latest effective number of holdings (1/Herfindahl index, from the most recent rebalance) across all fifteen funds. Equity/combined figures are annualised on 252 trading days and crypto on 365, matching each panel's own calendar.

Table 1. Out-of-sample performance and latest effective number of holdings, 2021-2023 (crypto: 2020-2023). Full turnover and concentration diagnostics are in Appendix A.

Fund	Ann. return	Ann. vol.	Sharpe	Max DD	Effective N
Equity Equal-Weight	13.2%	16.2%	0.82	-20.3%	50.0
Equity Min-Variance	6.2%	12.8%	0.49	-15.4%	9.2
Equity Max-Sharpe	10.7%	18.2%	0.59	-26.1%	3.1
Equity Risk-Parity	10.6%	14.6%	0.72	-18.5%	44.6
Equity Mean-CVaR	3.1%	13.7%	0.23	-17.0%	9.1
Crypto Equal-Weight	74.9%	81.0%	0.93	-81.6%	10.0
Crypto Min-Variance	82.0%	65.3%	1.26	-72.3%	2.1
Crypto Max-Sharpe	22.1%	77.3%	0.29	-89.3%	1.4
Crypto Risk-Parity	77.1%	78.7%	0.98	-79.5%	8.8
Crypto Mean-CVaR	54.6%	65.0%	0.84	-76.5%	1.9
Combined Equal-Weight	16.2%	21.3%	0.76	-28.7%	60.0
Combined Min-Variance	6.9%	13.4%	0.52	-15.6%	9.6
Combined Max-Sharpe	25.8%	24.7%	1.05	-26.3%	3.8
Combined Risk-Parity	14.4%	16.0%	0.90	-19.8%	51.1
Combined Mean-CVaR	4.7%	15.1%	0.31	-18.1%	9.1

Equal-weight is the standout result of this comparison, not an afterthought. In the equity universe it posts the highest Sharpe ratio of the five methods, 0.82, ahead of Risk-Parity (0.72), Max-Sharpe (0.59), Min-Variance (0.49), and Mean-CVaR (0.23), and its terminal growth of $1.43 also leads every other equity fund - no estimation of means, covariances, or tail risk beats doing nothing clever here, exactly the pattern DeMiguel, Garlappi and Uppal (2009) document. However, the ranking is not universal. In crypto, Min-Variance leads (Sharpe 1.26) with Equal-Weight third (0.93). In the combined universe, Max-Sharpe leads (1.05) with Equal-Weight fourth (0.76). Whether a naive benchmark or an estimated method wins depends on the asset universe, and reporting all five honestly is what makes that visible.

Crypto Max-Sharpe is the weakest fund in the whole comparison despite crypto's high average return. It has a Sharpe ratio of 0.29 and an 89.3% maximum drawdown, against Crypto Min-Variance's 1.26 and 72.3% built from the same ten coins. High average returns in an asset class do not guarantee that every optimisation method built on it performs well. Concentration and estimation error in expected returns (Michaud, 1989) separate the two crypto funds sharply.

Mean-CVaR's tail protection is real but narrow. Its maximum drawdown is the second shallowest of all five methods in all three universes, ahead of Max-Sharpe (-17.0% vs -26.1% Equity, -18.1% vs -26.3% Combined, -76.5% vs -89.3% Crypto) and of Risk-Parity and Equal-Weight throughout. In every universe, however, Min-Variance beats it on both dimensions at once. A shallower drawdown (-15.4% vs -17.0% Equity, -15.6% vs -18.1% Combined, -72.3% vs -76.5% Crypto) and a higher Sharpe ratio (0.49 vs 0.23, 0.52 vs 0.31, 1.26 vs 0.84). Minimising variance alone already controls tail risk about as well as minimising CVaR directly, leaving little room for a dedicated tail-risk objective to add value. Estimation error is the likely reason. The CVaR objective is fitted to the worst 5% of a 252-day window, roughly 13 observations, far fewer effective data points than a covariance matrix uses, so it generalises worse out of sample. The return floor becomes a cost but not a trade-off that paid off - Equity Mean-CVaR earns 3.1% annualised, the lowest of any equity fund, without any drawdown advantage over Min-Variance.

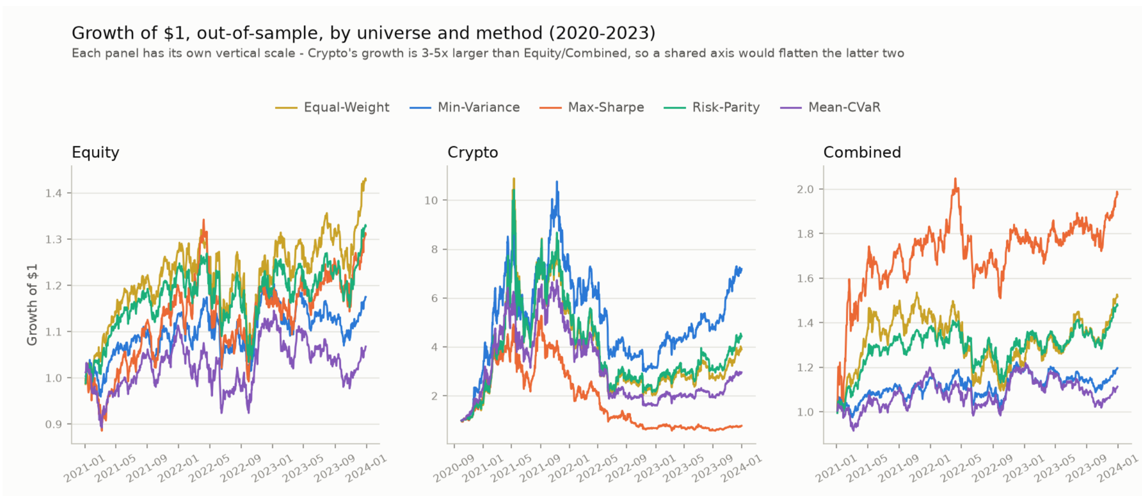

Figure 1. Growth of $1 across the fifteen base funds, shown by asset universe and optimisation method, out-of-sample 2020-2023.

Figure 1 shows a large spread in terminal wealth even within the crypto-only universe. One dollar grows to $7.19 under Min-Variance and $4.43 under Risk-Parity. However, under Max-Sharpe, despite there is a bull run in 2021, it ends at $0.77. Mean-CVaR sits between the two at $2.96. The combined panel compounds more steadily, with Max-Sharpe ending at $1.98, roughly double the equity-only funds, which cluster between $1.07 (Mean-CVaR) and $1.43 (Equal-Weight). High terminal wealth in the crypto panel therefore reflects the optimisation method as much as the asset class.

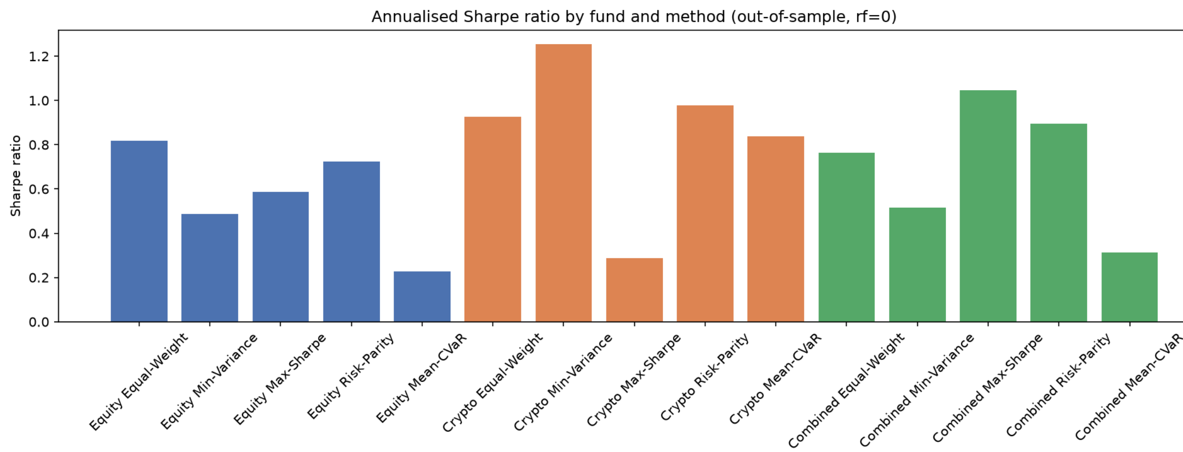

Figure 2. Annualised Sharpe ratio by fund and method, out-of-sample, rf = 0.

Figure 2 shows that optimisation method matters more than asset class for risk-adjusted performance. Crypto Minimum-Variance posts the highest Sharpe ratio overall at 1.26, but Crypto Max-Sharpe, built from the same ten coins, manages only 0.29 - the widest within-universe gap in the sample. Among the equity and combined products, Combined Max-Sharpe achieves the strongest result at 1.05, while Mean-CVaR is the weakest estimated method in both non-crypto universes, dominated by Min-Variance on drawdown as well as Sharpe ratio, so its weak result is not compensated by better tail protection.

2.2 Fund fact sheet: Combined Max-Sharpe

The Combined Max-Sharpe fund began its out-of-sample period on 4 January 2021, generating an annualised return of 25.8%, annualised volatility of 24.7%, a Sharpe ratio of 1.05, and a maximum drawdown of 26.3%. At the final rebalance on 1 December 2023, its non-zero holdings were GE 45.3%, NVDA 18.7%, BTC-USD 10.4%, ADBE 7.7%, SO 7.2%, BCH-USD 4.4%, INTC 4.0%, and TTWO 2.3%. Although 60 assets were available, the latest effective number of holdings was only 3.8, reflecting mean-variance optimisation's sensitivity to estimation error in expected returns (Michaud, 1989).

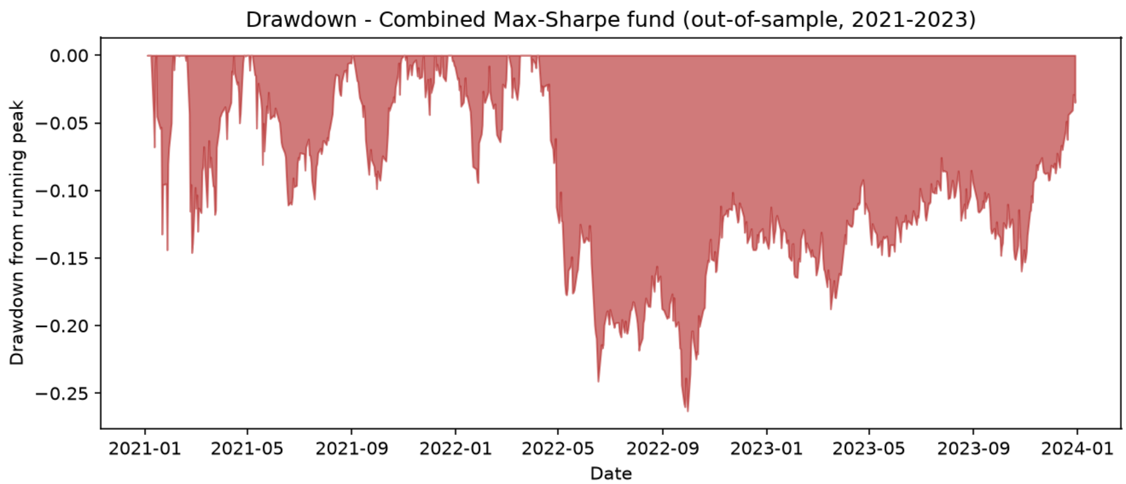

Figure 3. Out-of-sample drawdown of the Combined Max-Sharpe fund, 2021-2023.

Figure 3 shows that the fund's decline was not a brief shock. The fund peaks on 8 April 2022, falls 26.3% over the following six months to its trough on 30 September 2022, and does not recover to its prior peak for the remainder of the out-of-sample period, which ends in December 2023. A Sharpe ratio of 1.05 therefore compresses a materially longer and deeper drawdown than the summary statistic alone suggests.

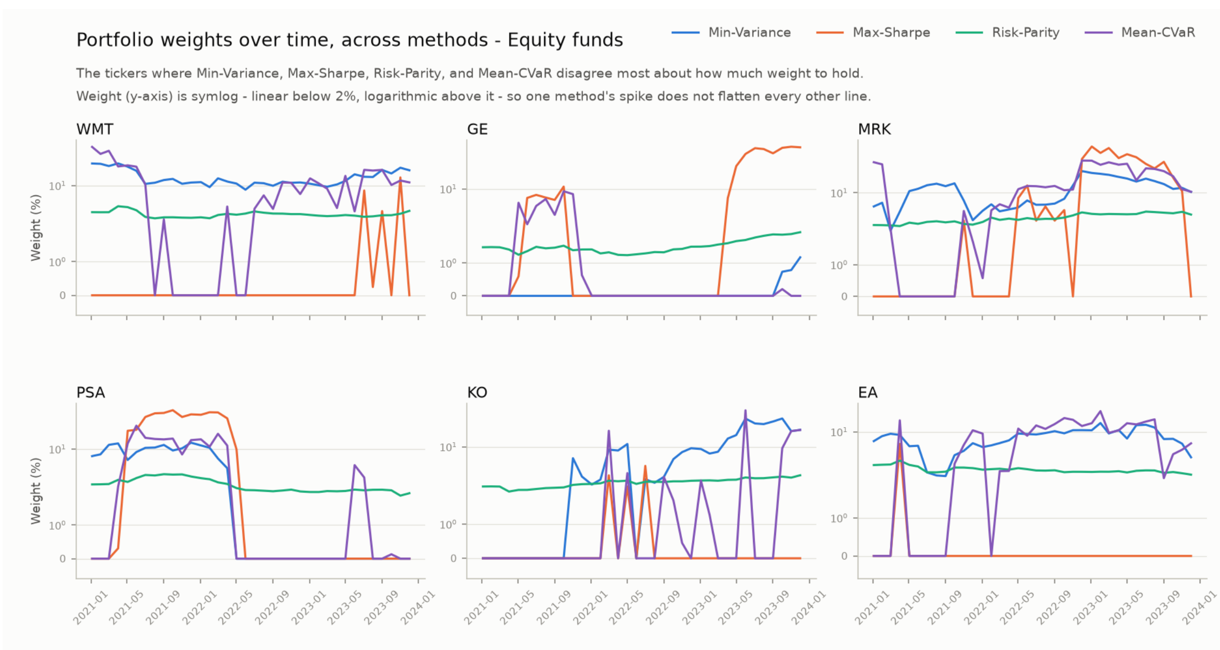

Figure 4. Equity portfolio weights over time, across minimum variance, maximum Sharpe, risk parity, and mean-CVaR, for the six tickers where the four methods disagree most. The y-axis is symlog (linear below 2%, logarithmic above it), since one method's weight in a single ticker can spike far above the others on a linear scale.

Figure 4 shows risk-parity weights staying relatively stable and dispersed, while maximum-Sharpe weights move more sharply and occasionally exceed 40-60% in a single stock. Minimum-variance and mean-CVaR weights generally sit between the two and often track each other closely, consistent with Table 1's similar effective-N holdings for both methods. Investors in Combined Max-Sharpe - the strongest historical risk-adjusted performer among the estimated methods - would still need to tolerate concentration, model instability, and a drawdown of roughly one quarter of capital.

3. Standalone sector news-sentiment index

3.1 Finance-adapted text model

The news data are headlines, not full articles, so sentiment here is a noisy, compressed proxy for the information environment. Text is kept in its original casing and punctuation because VADER uses capitalisation, emphasis, negation, and punctuation as part of its rule based score (Hutto & Gilbert, 2014). Each headline h is scored with a compound score in [-1, 1], and every ticker k's headlines on day t are averaged to a ticker-day score:

$$s_{k,t} = mean\big(compound(h) \;\; for\; h\; in\; headlines(k,t)\big).$$

finVADER (Koráb, 2023) extends the standard VADER lexicon with two finance word lists, SentiBigNomics (Consoli, Barbaglia & Manzan, 2022) and Henry's list (Henry, 2008), aiming to reduce false neutrality where a word carries a clear financial meaning a general-purpose lexicon does not capture. This consistent with evidence that generic dictionaries misclassify ordinary financial and accounting vocabulary (Loughran & McDonald, 2011). Table 2 shows the measured effect: the proportion of exact-zero ticker-day scores falls from 23.0% under standard VADER to 5.9% under finVADER, while mean sentiment decreases from 0.113 to 0.072, indicating the extension changes both coverage and score calibration.

A self-built lexicon extension was also constructed and validated. Candidate finance terms were checked against the combined VADER and finVADER vocabulary (13,324 words) and revealed several asymmetric gaps. FinVADER covers “hawkish” but not “dovish”, “headwind” but not “tailwind”, “overbought” but not “oversold”.

The 13 missing candidates were all independently rated twice to estimated their meaning in financial news headline. They are only kept when both passes agreed in sign and were within 0.25 of each other on a -1 to +1 scale. There are two candidates, “deleveraging” and “derisking” failed in this check. Both can read as prudent discipline or as a symptom of distress depending on context and were dropped. The remaining 11 terms form the custom lexicon in Table 2.

Table 2. Standard VADER, finVADER, and a self-built 11-term lexicon extension, scored on the same 37,962 ticker-day headline observations, 2020-2023.

Model	Mean ticker-day sentiment	Exact-zero observations
VADER	0.113	23.0%
finVADER	0.072	5.9%
Custom (VADER + self-built lexicon)	0.114	22.8%

The custom lexicon barely moves the aggregate statistics. The mean sentiment is 0.114 and the exact zero rate is 22.8%, they are both close to plain VADER's 0.113 and 23.0%. In aggregate, eleven words cannot compete with finVADER's roughly 7,500 term extension. The value of the method is its disciplined, checkable rating process. finVADER, the more thoroughly validated of the two extensions, remains the production model for the sector index (Section 3.2) and the sentiment fusion (Section 4.1).

3.2 Index construction and interpretation

The sector index averages ticker scores equally within each sector, so no sector's reading is dominated by its most frequently covered company. For ticker-days without a new headline, instead of treating them as zero, the ticker's most recently observed score is carried forward, since a compound score of zero would incorrectly equate missing news with neutral news. Before a ticker's first headline it is excluded from the average altogether, rather than entered as a zero.

Sector g's index on day t averages that sector's N_g ticker scores:

$$I_{g,t} = \frac{1}{N_g} \sum_{k \in g} s_{k,t}.$$

Each sector series is then standardised against its own 2020-2023 mean and standard deviation:

$$z_{g,t} = \frac{I_{g,t} - \bar{I}_g}{\sigma_g}.$$

Raw finVADER scores are positive on 84.2% of all sector days, so an unstandardised index would read as "positive" almost every day and could not distinguish an unusual day from a normal one. Standardising makes a sector's greedy or fearful days readable relative to its own norm. The standardised series borrows fear and greed terminology from the market wide Fear & Greed Index published by CNN Business (CNN Business, n.d.), which likewise rescales a composite indicator so extreme readings are read relative to a normal range. The label is descriptive only. This index is built solely from finVADER headline z scores and shares no inputs or methodology with CNN's version. Standardising still does not make sectors comparable in absolute terms, since each is standardised against its own history, not a shared one.

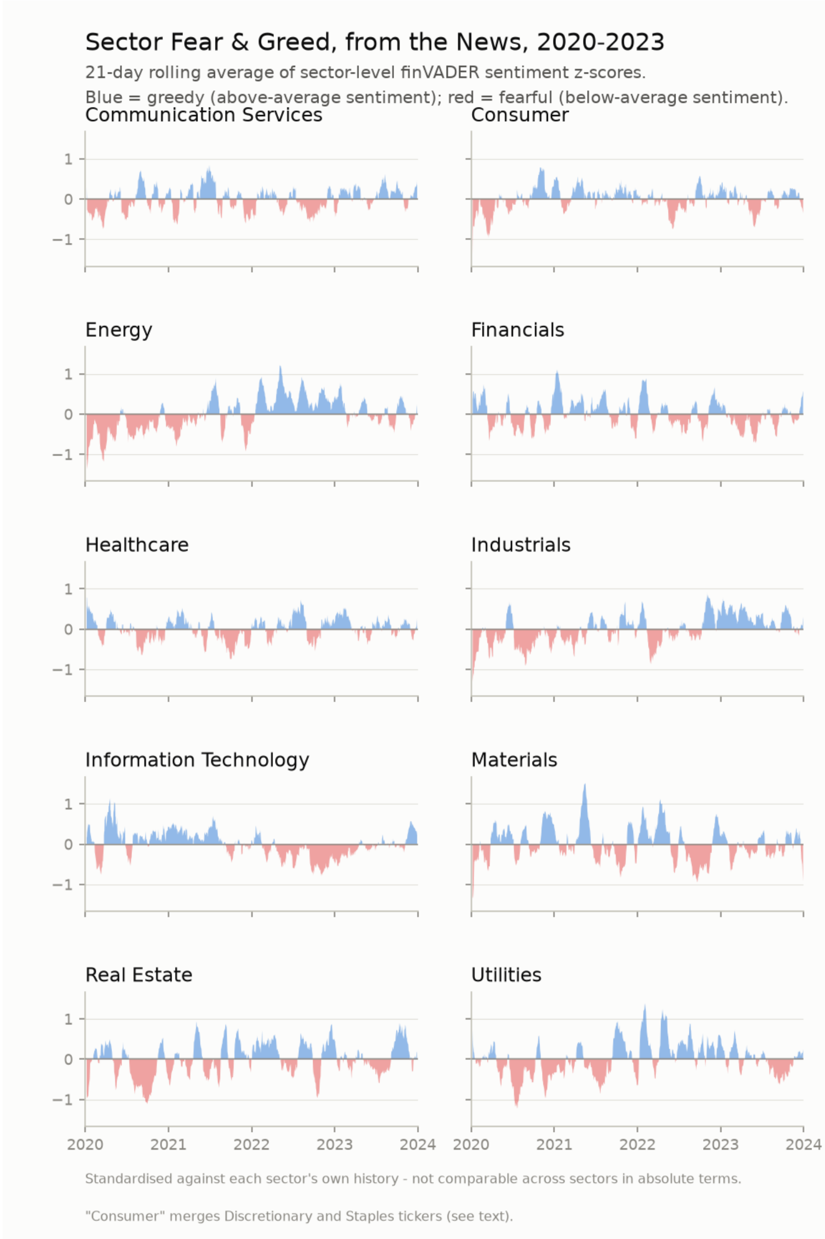

Figure 5. Sector fear and greed from the news, 2020-2023, 21-trading-day rolling average of finVADER z-scores. Blue = greedy (above-average sentiment); red = fearful (below-average sentiment).

Figure 5 lindicates both a common stress period and real cross sector differences. During the COVID crash window, Energy sentiment falls below its own norm, averaging a z-score of -0.72 (most fearful), while Real Estate and Technology stay mildly above their own norms even through the crash and remain at +0.39 and +0.23 (mildly greedy). Across the full 2020-2023 sample, Real Estate, Industrials, and Technology has the most positive average raw sentiment, Energy and Materials the least. Sectors such as Real Estate and Utilities which are thinly covered also show more abrupt movements. Therefore, the figure reflects both real news tone and coverage intensity.

4. Extensions and innovations

This project extends beyond the required minimum in several ways. Fifteen funds span three universes and five optimisation methods which includes an honest equal-weight benchmark and a mean-CVaR method solved as an exact linear program - well beyond the required single combined fund (Section 1.1), and equal-weight beats every estimated equity method out of sample (Section 2.1), reported because it is true, not because it flatters the estimated methods. Two optimiser-validation checks caught different silent convergence failures. An SLSQP scaling issue leaving 142 of 222 minimum-variance and risk-parity rebalances (64%) at equal weight despite reported success, and a non-smooth-objective issue affecting up to 13 of 36 mean-CVaR rebalances, fixed by reformulating the problem (Section 1.4). Concentration and turnover diagnostics are added to every fund, surfacing risk that raw Sharpe ratios alone would hide (Sections 2, 5.2). finVADER reduces exact-zero sentiment classifications from 23.0% to 5.9% (Section 3.1), and a self-built 11 term lexicon extension, rated in two independent passes and validated against finVADER, adds a further checkable comparison (Section 3.1). The following section presents the sentiment fusion in detail.

4.1 Look-ahead-safe tilt design

The fusion extension tilts the Equity Max-Sharpe fund's base weight for each stock i in proportion to that stock's own recent sentiment, using a tilt strength λ:

$$\tilde{w_i} = w_i^{base}\ (1 + \lambda s_i).$$

The tilted weights are then floored at zero and renormalised so the fund stays long-only and fully invested:

$$w_i = \frac{max(\tilde{w_i},0)}{\sum_j max(\tilde{w_j},0)}.$$

Here s_i is ticker i's most recently observed raw finVADER score strictly before the rebalance date, at least one trading day's lag, so a Saturday or Monday headline is first usable for Tuesday's trade and the tilt is look-ahead safe. If a ticker does not have a prior sentiment observation, it gets no tilt. The main specification uses λ = 0.50.

Table 3. Equity Max-Sharpe fund performance before and after the 0.50 sentiment tilt, out-of-sample 2021-2023.

Fund	Return	Vol.	Sharpe	Max DD	Turnover	Max wt.	Effective N
Equity Max-Sharpe	10.69%	18.23%	0.586	-26.12%	34.2%	50.4%	3.13
+ Sentiment tilt	10.68%	18.28%	0.584	-26.48%	34.5%	52.4%	2.98

On every reported risk adjusted dimension, the main tilt is slightly negative. In Table 3, annualised return decreases from 10.69% to 10.68%, volatility rises from 18.23% to 18.28%, and the Sharpe ratio declines from 0.586 to 0.584. Maximum drawdown deepens from -26.12% to -26.48%. This is economically small, but consistent in direction. The signal is not enough to offset the extra portfolio distortion. Concentration also increases, the latest maximum weight increase from 50.4% to 52.4% and the effective number of holdings falling from 3.13 to 2.98.

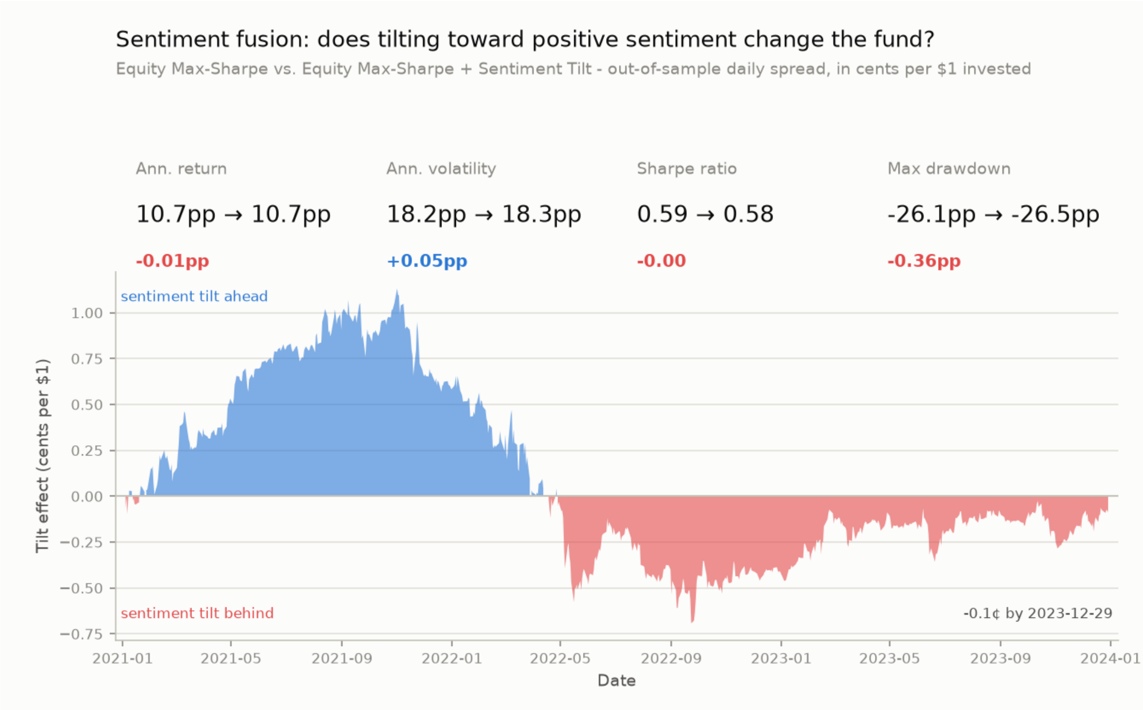

Figure 6. Cumulative effect of the sentiment tilt (λ = 0.50) relative to the Equity Max-Sharpe base fund, cents per $1 invested, out-of-sample.

Figure 6 indicates that the tilt's advantage over the base fund has peaked on 1 November 2021 for 1.13 cents each dollar. Performance reversed across 2022, reaching its largest deficit of -0.69 cents on 23 September 2022. By the end of the sample, it narrows to -0.09 cents. The tilt adds value during the rising sentiment regime it was designed for. However, it gives most of it back once sentiment and price momentum diverge in 2022.

4.2 Robustness across tilt strengths

It cannot be classified whether the result is representative or an accident of one parameter through choosing a single tilt strength. Therefore, the fusion uses three additional predetermined strengths, 0.25, 1.00, and 0 (no tilt) to rerun. These three index are chose before looking at any result.

Table 4. Pre-specified sentiment-tilt robustness test across four values of λ.

Tilt λ	Ann. return	Ann. vol.	Sharpe	Max DD	Turnover
0.00	10.69%	18.23%	0.586	-26.12%	34.2%
0.25	10.68%	18.25%	0.585	-26.30%	34.3%
0.50	10.68%	18.28%	0.584	-26.48%	34.5%
1.00	10.67%	18.34%	0.581	-26.81%	35.1%

As λ rises from 0 to 1, annualised return, Sharpe ratio, and maximum drawdown all worsen monotonically while volatility and turnover rise. This means that the roughly neutral result at λ = 0.50 is not an accident of that one value. Stronger reliance on headline sentiment consistently reduces this fund's performance across the tested range. There are few explanations for this. First, news tone identified by finVADER may already be reflected in market prices before the monthly rebalance. Second, carried-forward sentiment may go stale and dominate days with little news. Third, applying the tilt to an already concentrated Maximum-Sharpe portfolio may amplify estimation error. Overall, the sector index is a useful independent analysis, but the current tilt is not yet an investable enhancement.

5. Streamlit application and investor journey

5.1 Target user and workflow

SentiFolio is facing to retail investors seeking systematic exposure to equities and cryptocurrencies. They don't need to select individual securities or estimating covariance matrices themselves. The journey begins in the Funds tab, this is where investors compare return, volatility, Sharpe ratio, drawdown, turnover, and concentration across the full product range. Then, they can open a chosen fund's fact sheet to review its growth of $1 and drawdown through time and check its current target holdings. This allows users to check whether the concentration and loss profile align with their risk tolerance.

The Sentiment tab is an independent layer that support decision making. A user can select a specific sector and view its standardised finVADER history and compare it with sector's own norm.

"Build a portfolio" lets the investor assign percentages across the sixteen available strategies. It includes fifteen base funds and a sentiment-tilted equity fund. It calculates the blended portfolio's historical return, volatility, Sharpe ratio, drawdown, and growth. Dates are selected when every selected fund is live, so the period before a fund's launch is never treated as a zero return day. Only when every selected fund is crypto only does the blend annualise on 365 days. In contrast, when equity or combined fund is included, performance is annualised using 252 days. This is because the intersected sample follows the equity trading calendar.

5.2 Implementation choices

The deployed application only reads CSV artefacts that are precomputed from the results directory. Opening the page never downloads raw data, runs NLTK, scores headlines, or reruns optimisation. This keeps the app responsive on a basic Streamlit Community Cloud machine and ensures the report and the app describe the same reproducible outputs. The repository contains the code, results artefacts, requirements, and AI workflow documentation, while raw data and API credentials are excluded from version control and loaded at runtime through src/data_access.py.

The interface also indicates both average one-way turnover and the effective number of holdings. These measures exceed the minimum fact sheet because two funds with similar Sharpe ratios can carry very different implementation risk. Risk parity combines low turnover with broad diversification. However, maximum-Sharpe is concentrated and rebalances more aggressively.

The app is still a prototype. It does not execute trades, considerate investor tax circumstances, or provide personal financial advice.

6. Critical reflection and recommendations

6.1 What worked

Fifteen funds are generated through one common walk-forward framework. The app exposes the same returns, holdings, and metrics used in the report. Two rounds of optimiser validation improved the credibility of every result depend on solver. First round caught the SLSQP scaling problem for minimum-variance/risk-parity. Second round identified the non-smooth objective problem for mean-CvaR. Each round checked whether the solver had made an efficient search.

The project has added Equal-Weight method and produced a striking result. The naive 1/N benchmark defeated every estimated equity method out of sample. This result matches the finding of DeMiguel, Garlappi and Uppal (2009). Another valuable extension will be the finance lexicon. It can significant decrease the false-neutral sentiment classification and generates a sector index with distinct, interpretable time-series behaviour.

6.2 What did not work as expected

The sentiment fusion did not improve the base fund. The robustness table shows that the larger the tilts strength is, the worse the outcome become. Therefore, this extension should not be treated as evidence that headline sentiment increases returns. The outcome exposes both weaknesses in the signal and the portfolio it is applied to. Headline tone may decay faster than a monthly rebalance can capture. Forward-filled sentiment can go stale while a concentrated maximum-Sharpe base amplifies a small signal error. The zero-transaction-cost assumption also likely overstates implementable performance for higher-turnover funds - Equity Max-Sharpe already turns over 34.2% of its book per rebalance before any tilt - and crypto's unusually strong 2021-2023 run makes the strongest historical results the least likely to repeat.

6.3 Three concrete real world recommendations

First, control concentration and estimation risk directly. A 20% single-asset cap should be added before optimisation and apply shrinkage processing to estimated means and covariances. The current Combined Max-Sharpe portfolio places 45.3% in GE with an effective number of holdings of only 3.8. Meanwhile, this report has shown that Equal-Weight beat every estimated equity method, and this supports simpler and more robust construction. The revised fund should be evaluated in comparison with the unconstrained version while examine Sharpe ratio, drawdown, turnover, and concentration together at the same time.

Second, redesign sentiment as a decaying, higher-frequency signal. The current unlimited carry-forward could be replaced with a maximum horizon of five trading days, combined with an exponential decay rule. Furthermore, weekly sentiment rebalancing could be tested instead of monthly rebalancing. A realistic transaction costs could be included too. This directly addresses the possibility that headline information is short-lived and avoids an old headline indefinitely influencing a portfolio.

Third, move from a fixed lexicon tilt to validated event and topic features. Separate earnings, guidance, litigation, merger, and macroeconomic headlines. Then, test whether each category has different predictive value. During this process, a training/validation split should be used to develop lexicon and parameter and leave the final period unchanged for evaluation. This not only remain the interpretability of finVADER but also reducing data snooping. At the same time, it also identifies whether only specific news types are tradable.

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Appendix A - Supplementary diagnostics

A1. Full performance and product diagnostics

Table A1. Full performance metrics, including implementation and concentration diagnostics, for all fifteen base funds and the sentiment-tilted fund.

Fund	Return	Vol.	Sharpe	Max DD	Turnover	Max wt.	HHI	Eff. N	Ann.
Equity Equal-Weight	13.2%	16.2%	0.82	-20.3%	0.0%	2.0%	0.020	50.0	252
Equity Min-Variance	6.2%	12.8%	0.49	-15.4%	15.1%	18.8%	0.108	9.2	252
Equity Max-Sharpe	10.7%	18.2%	0.59	-26.1%	34.2%	50.4%	0.320	3.1	252
Equity Risk-Parity	10.6%	14.6%	0.72	-18.5%	2.0%	4.2%	0.022	44.6	252
Equity Mean-CVaR	3.1%	13.7%	0.23	-17.0%	33.5%	18.7%	0.110	9.1	252
Crypto Equal-Weight	74.9%	81.0%	0.93	-81.6%	0.0%	10.0%	0.100	10.0	365
Crypto Min-Variance	82.0%	65.3%	1.26	-72.3%	13.1%	56.5%	0.474	2.1	365
Crypto Max-Sharpe	22.1%	77.3%	0.29	-89.3%	32.5%	82.1%	0.698	1.4	365
Crypto Risk-Parity	77.1%	78.7%	0.98	-79.5%	1.8%	18.8%	0.114	8.8	365
Crypto Mean-CVaR	54.6%	65.0%	0.84	-76.5%	13.0%	61.3%	0.525	1.9	365
Combined Equal-Weight	16.2%	21.3%	0.76	-28.7%	0.0%	1.7%	0.017	60.0	252
Combined Min-Variance	6.9%	13.4%	0.52	-15.6%	17.5%	18.5%	0.104	9.6	252
Combined Max-Sharpe	25.8%	24.7%	1.05	-26.3%	36.9%	45.3%	0.266	3.8	252
Combined Risk-Parity	14.4%	16.0%	0.90	-19.8%	2.2%	3.8%	0.020	51.1	252
Combined Mean-CVaR	4.7%	15.1%	0.31	-18.1%	35.1%	21.1%	0.110	9.1	252
Equity Max-Sharpe + Sentiment Tilt	10.7%	18.3%	0.58	-26.5%	34.5%	52.4%	0.336	3.0	252